Weil Conjectures

Appendix B: Abelian varieties

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B Abelian varieties

Two steps of the main proof come down to the torsion points of a Jacobian. By the Kummer sequence, $H^1(C, \mu_n)$ of a curve is the n-torsion of its Jacobian. The first step, the finiteness of H^1 needed for proper base change, is therefore a count of torsion points. The second step is the finiteness of geometric monodromy in rank one, where purity begins. It needs Frobenius to fix no class in H^1 of a curve over $\mathbf{F}_q$ (Proposition 14.3*).

For a complex torus $\mathbf{C}^g/\Lambda$ the count is trivial: the n-torsion is $\frac{1}{n}\Lambda/\Lambda \simeq (\mathbf{Z}/n)^{2g}$. For an abelian variety A of dimension g over an arbitrary field k, however, there is no lattice, and the torsion points must be counted algebraically. This is what we do: we prove that for n invertible in k the group scheme $A[n]$ is finite étale of rank n^{2g} (Proposition B.12). It follows that the Tate module $T_\ell A = \varprojlim_m A[\ell^m](\bar{k})$ is free of rank $2g$ over $\mathbf{Z}_\ell$ (Proposition B.14). Moreover, over $\mathbf{F}_q$, no power of Frobenius has a nonzero fixed vector on $T_\ell A$ (Proposition B.15). The reason is short: a fixed vector would be a compatible system of ℓ-power torsion points of unbounded order in the finite group $A_0(\mathbf{F}_{q^m})$.

The count rests on one formula. If $\mathcal{N}$ is a symmetric ample invertible sheaf on A, then

$$[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$$

(Proposition B.11). On an elliptic curve with $\mathcal{N} = \mathcal{O}(e)$ this says that the divisor $[n]^*(e)$ is linearly equivalent to $n^2(e)$. When n is prime to the characteristic, $[n]$ is étale and $[n]^*(e)$ is the reduced divisor $E[n]$. Comparing degrees then gives $\#E[n] = n^2$.

In general the formula is used twice. First, it forces $[n]$ to have finite fibres. Indeed, $\mathcal{N}^{\otimes n^2}$ has positive degree on every complete curve in A, while $[n]^*\mathcal{N}$ is trivial on a curve contracted by $[n]$. Second, we compute the Euler characteristics $\chi(A, [n]^*\mathcal{N}^{\otimes m})$ in two ways, and comparing their leading coefficients gives $\deg[n] = n^{2g}$.

The formula itself comes from the theorem of the cube (Theorem B.6). Following [Stacks, 0BEZ], we prove the theorem of the cube together with the rigidity lemma. As a by-product, from the rigidity lemma we deduce that A is commutative; see also [Mil86a].

In this appendix all group schemes are of finite type over a field k of arbitrary characteristic.

B.1 The theorem of the cube

The theorem of the cube says that an invertible sheaf $\mathcal{L}$ on $X \times Y \times Z$, with X, Y complete varieties and Z connected, that is trivial on the three slices through a point (x, y, z) is trivial. To prove it, we view $\mathcal{L}$ as a family of invertible sheaves on $X \times Y$ parametrized by Z. Following [Stacks, 0BEZ], we then consider the set of points of Z over which the sheaf is trivial. This set is closed, by semicontinuity. It is also open, by a computation of H^0 and H^1 near such a point. Since it contains z and Z is connected, it is all of Z. The openness uses the triviality on the two slices $x \times Y \times Z$ and $X \times y \times Z$, and with only two factors it fails.

Example B.1 (Two factors do not suffice)**.** Let E be an elliptic curve over an algebraically closed field, with origin O, and let $\Delta \subset E \times E$ be the diagonal. On $E \times E$ put

$$\mathcal{L} = \mathcal{O}_{E\times E}(\Delta - E \times \{O\} - \{O\} \times E).$$

Consider the slice $E \times \{t\}$. It meets the diagonal in the point t, and the divisor $\{O\} \times E$ meets it in the point O. The divisor $E \times \{O\}$, on the other hand, is pulled back from the second factor, so it contributes the trivial sheaf for every t. Hence the restriction is $\mathcal{O}_E(t - O)$. By symmetry the restriction to $\{t\} \times E$ is also $\mathcal{O}_E(t - O)$. In particular $\mathcal{L}$ is trivial on both slices through (O, O).

However, $\mathcal{L}$ is not trivial, because for $t \neq O$ the sheaf $\mathcal{O}_E(t - O)$ is nontrivial. Indeed, a rational function with divisor $t - O$ would be a map $E \to \mathbf{P}^1$ of degree one, which is impossible in genus one. The set of t over which the fibre is trivial is the point O. It is closed, as Lemma B.2 below predicts, but not open.

Lemma B.2 (Triviality on fibres)**.** *Let P be a proper geometrically integral k-scheme, let T be a k-scheme of finite type, let $f : P \times T \to T$ be the projection, and let $\mathcal{L}$ be an invertible sheaf on $P \times T$. For $t \in T$ write $P_t = P \times_k \kappa(t)$ and $\mathcal{L}_t = \mathcal{L}|_{P_t}$.*

1. *$\mathcal{L}_t$ is trivial if and only if $H^0(P_t, \mathcal{L}_t) \neq 0$ and $H^0(P_t, \mathcal{L}_t^{-1}) \neq 0$.*
2. *The set of $t \in T$ such that $\mathcal{L}_t$ is trivial is closed.*

Proof. The proof uses only that P_t is proper and integral with $H^0(P_t, \mathcal{O}_{P_t}) = \kappa(t)$, which holds by [Stacks, 0BUG] because P is proper and geometrically integral.

For (1), a trivial sheaf and its inverse both have the nonzero section 1, so only the converse needs an argument. On $\mathbf{P}^1$, for instance, $\mathcal{O}(d)$ and $\mathcal{O}(-d)$ both have nonzero sections only when $d = 0$, and the general argument is the same. Let s and s' be nonzero sections of $\mathcal{L}_t$ and $\mathcal{L}_t^{-1}$. Since P_t is integral, the product ss' is a nonzero element of $H^0(P_t, \mathcal{O}) = \kappa(t)$, hence a unit. Since ss' would vanish wherever s does, the section s vanishes nowhere, and it trivializes $\mathcal{L}_t$.

For (2), note that by (1) the set in question is the intersection of the loci where $\dim H^0(P_t, \mathcal{L}_t)$ and $\dim H^0(P_t, \mathcal{L}_t^{-1})$ are positive. These loci are closed because the functions $t \mapsto \dim H^0(P_t, \mathcal{L}_t^{\pm 1})$ are upper semicontinuous [Stacks, 0BDN], as $\mathcal{L}$ is flat over T and f is proper. ∎

Part (2) is the closed half of the theorem of the cube. If the sheaf is trivial over the members of an open cover of T, then the evaluation map identifies it with a pullback from T, since sections along the proper factor P are constant.

Corollary B.3 (Triviality over an open cover of the base)**.** *With the notation of Lemma B.2, suppose that $\mathcal{L}$ is trivial on $P \times U_\alpha$ for an open cover (U_α) of T. Then $f_*\mathcal{L}$ is invertible and the evaluation map $f^* f_*\mathcal{L} \to \mathcal{L}$ is an isomorphism.*

Proof. Both assertions are local on T, so we may replace T by one of the U_α and assume that $\mathcal{L} \simeq \mathcal{O}_{P\times T}$. Since $H^0(P, \mathcal{O}_P) = k$ [Stacks, 0BUG], and T is flat over k, flat base change gives $f_*\mathcal{O}_{P\times T} = \mathcal{O}_T$. Hence the chosen trivialization identifies $f_*\mathcal{L}$ with the invertible sheaf $\mathcal{O}_T$. Under the same identification the evaluation map $f^* f_*\mathcal{L} \to \mathcal{L}$ becomes the identity of $\mathcal{O}_{P\times T}$. ∎

The open half of the theorem is the substantial step. It uses the fact that H^1 of the structure sheaf of a product is the sum of the H^1 of the factors.

Lemma B.4 (H^1 of the structure sheaf of a product)**.** *Let K be a field, let P and Q be proper K-schemes with $H^0(P, \mathcal{O}_P) = H^0(Q, \mathcal{O}_Q) = K$, and let $p_0 \in P(K)$ and $q_0 \in Q(K)$. Restriction to the slices $P \times q_0$ and $p_0 \times Q$ is an isomorphism*

$$H^1(P \times Q, \mathcal{O}) \longrightarrow H^1(P, \mathcal{O}) \oplus H^1(Q, \mathcal{O}).$$

Proof. By the Künneth formula [Stacks, 0BED], and since H^0 of $\mathcal{O}$ is K on both factors, the source is $\mathrm{pr}_1^* H^1(P, \mathcal{O}) \oplus \mathrm{pr}_2^* H^1(Q, \mathcal{O})$. Restriction to $P \times q_0$ is the identity on the first summand. It kills the second, because on the second summand it factors through H^1 of the point q_0. Restriction to $p_0 \times Q$ does the opposite. ∎

Lemma B.5 (Triviality near a fibre)**.** *Let P and Q be proper geometrically integral k-schemes with points $p_0 \in P(k)$ and $q_0 \in Q(k)$, let T be a k-scheme of finite type, and let $\mathcal{L}$ be an invertible sheaf on $P \times Q \times T$ whose restrictions to $p_0 \times Q \times T$ and to $P \times q_0 \times T$ are trivial. If $\mathcal{L}$ is trivial on the fibre over a point $t \in T$, then $\mathcal{L}$ is trivial on $P \times Q \times U$ for an open neighbourhood U of t.*

Proof. We want a section of $\mathcal{L}$ over $X \times U$, where $X = P \times Q$, that does not vanish on the fibre over t. Its zero locus then misses a neighbourhood of that fibre. Such a section exists as soon as $f_*\mathcal{L}$ is free of rank one near t, with formation compatible with passage to the fibre.

To get this, we represent $Rf_*\mathcal{L}$, and the pushforwards of the restrictions of $\mathcal{L}$ to the two slices, by complexes of free modules whose differentials vanish at t. By the Künneth formula, restriction from H^1 of the fibre to H^1 of the two slices is an isomorphism. This forces the first differential of $Rf_*\mathcal{L}$ to vanish after shrinking T. Then H^0 of $Rf_*\mathcal{L}$ is free of rank one, as we need.

Setup. Since $X_{\bar{k}} = P_{\bar{k}} \times Q_{\bar{k}}$ is a product of integral varieties, X is geometrically integral [Stacks, 05P3]. We may assume that $T = \operatorname{Spec} R$ is affine, and we let $\mathfrak{p}$ be the prime of t. Below, "shrinking T" means replacing R by R_g for some $g \notin \mathfrak{p}$; everything already constructed survives this. Consider the projections

$$f : X \times T \to T, \qquad f_1 : P \times T \to T, \qquad f_2 : Q \times T \to T,$$

and the restrictions $\mathcal{L}_1$ and $\mathcal{L}_2$ of $\mathcal{L}$ to $P \times q_0 \times T$ and $p_0 \times Q \times T$, which are trivial. The complexes $E = Rf_*\mathcal{L}$ and $E_i = Rf_{i*}\mathcal{L}_i$ are perfect, and their formation commutes with base

change [Stacks, 0B91]. Restriction of sections gives a map $E \to E_1 \oplus E_2$, also compatible with base change. Since T is affine, $H^0(X \times T, \mathcal{L}) = H^0(E)$ and likewise for E_i.

Minimal complexes. A perfect complex over a local ring can be represented by a complex of free modules whose differentials vanish modulo the maximal ideal; the ranks are then the dimensions of the cohomology of the fibre. We use this to read off the ranks from the fibre over t, where everything is trivial. To do so, put $h^j = \dim H^j(X_t, \mathcal{O})$ and $h_i^j = \dim H^j$ of $\mathcal{O}$ on the fibres P_t and Q_t. Since $\mathcal{L}$ is trivial on X_t and the $\mathcal{L}_i$ are trivial, $E \otimes^{\mathbf{L}}_R \kappa(t)$ and $E_i \otimes^{\mathbf{L}}_R \kappa(t)$ have cohomology of dimensions h^j and h_i^j, with $h^0 = h_i^0 = 1$. After shrinking T we may therefore represent E and E_i by complexes of finite free modules

$$R \xrightarrow{d} R^{h^1} \to R^{h^2} \to \cdots, \qquad R \xrightarrow{d_i} R^{h_i^1} \to R^{h_i^2} \to \cdots$$

whose differentials vanish modulo $\mathfrak{p}$ [Stacks, 0BCD]. We represent the map $E \to E_i$ by a map of complexes [Stacks, 064B], with components $a_i : R \to R$ in degree 0 and $b_i : R^{h^1} \to R^{h_i^1}$ in degree 1. Since the differentials vanish modulo $\mathfrak{p}$, the map $b_i \otimes \kappa(t)$ is the restriction map on H^1 of the fibre over t.

We want to show that $d = 0$ after shrinking T, since then $H^0(E) = \ker d = R$ is free of rank one. The maps b_i will let us transfer this question to the d_i, which are easier because $\mathcal{L}_i$ is trivial.

The differentials d_i vanish. Fix a trivialization $\mathcal{L}_1 \simeq \mathcal{O}$, and let $u \in H^0(E_1) = \ker d_1 \subset R$ be the section corresponding to 1. Its image in $H^0(E_1 \otimes^{\mathbf{L}} \kappa(t)) = H^0(P_t, \mathcal{O})$ is $1 \neq 0$, so $u \notin \mathfrak{p}$. Since $d_1(u) = u\, d_1(1) = 0$, after inverting u we get $d_1 = 0$. The same argument gives $d_2 = 0$.

The differential d vanishes. This step uses the two slices together. If the map (b_1, b_2) is an isomorphism, then d is determined by d_1 and d_2, and this condition can be checked on the fibre. The map $(b_1, b_2) \otimes \kappa(t)$ is restriction from $H^1(P_t \times Q_t, \mathcal{O})$ to the slices $P_t \times q_0$ and $p_0 \times Q_t$. It is an isomorphism by Lemma B.4, and in particular $h^1 = h_1^1 + h_2^1$. Hence the square matrix (b_1, b_2) has determinant outside $\mathfrak{p}$. After inverting this determinant, (b_1, b_2) is an isomorphism. Since the maps form a map of complexes, we have $b_i \circ d = d_i \circ a_i = 0$, and therefore $d = 0$.

Conclusion. A generator of H^0 gives the trivialization. We have $H^0(X \times T, \mathcal{L}) = H^0(E) = \ker d = R$, and its generator σ maps to a generator of $H^0(E \otimes^{\mathbf{L}} \kappa(t)) = H^0(X_t, \mathcal{L}_t)$. Since $\mathcal{L}_t \simeq \mathcal{O}_{X_t}$ and $H^0(X_t, \mathcal{O}) = \kappa(t)$, the section σ vanishes nowhere on X_t. Its zero locus is closed in $X \times T$ and disjoint from X_t. Since f is proper, its image in T is a closed set not containing t. On the complement U of this image, σ trivializes $\mathcal{L}$. ∎

With the closed and the open half in hand, it remains to glue the local trivializations.

Theorem B.6 (Theorem of the cube)**.** *Let X and Y be proper geometrically integral k-schemes with points $x \in X(k)$ and $y \in Y(k)$, let Z be a connected k-scheme of finite type with a point $z \in Z(k)$, and let $\mathcal{L}$ be an invertible sheaf on $X \times Y \times Z$. If the restrictions of $\mathcal{L}$ to $x \times Y \times Z$, $X \times y \times Z$ and $X \times Y \times z$ are trivial, then $\mathcal{L}$ is trivial.*

Proof. The local trivializations are glued by Corollary B.3, which shows that $\mathcal{L}$ is pulled back from $Y \times Z$; the slice $x \times Y \times Z$ then identifies the sheaf on $Y \times Z$.

Let $Z_0 \subset Z$ be the set of points over which $\mathcal{L}$ is trivial on the fibre of $p : X \times Y \times Z \to Z$. It contains z. It is closed by Lemma lemma-triviality-fibres applied to $X \times Y$, which is proper and geometrically integral [Stacks, 05P3]. It is open by Lemma B.5. Since Z is connected, $Z_0 = Z$. Now Lemma B.5 gives an open cover of Z over whose members $\mathcal{L}$ is trivial.

Let $\pi : X \times Y \times Z \to Y \times Z$ be the projection and σ its section through x. We apply Corollary B.3 with $P = X$, $T = Y \times Z$ and the open cover by the products of Y with these opens. It shows

that $\mathcal{M} = \pi_*\mathcal{L}$ is invertible and that $\pi^*\mathcal{M} \to \mathcal{L}$ is an isomorphism. Hence $\mathcal{M} \simeq \sigma^*\pi^*\mathcal{M} \simeq \sigma^*\mathcal{L}$. This is the restriction of $\mathcal{L}$ to $x \times Y \times Z$, which is trivial by hypothesis. We conclude that $\mathcal{L} \simeq \pi^*\mathcal{M}$ is trivial. ■

The second tool, which we use to show that an abelian variety is commutative (Lemma B.9), is a rigidity statement. It says that a family of maps from a proper variety that is constant on one member is constant on all nearby members.

Lemma B.7 (Rigidity)**.** *Let X be a proper geometrically integral k-scheme, let Y be a k-scheme of finite type, let Z be a k-scheme, and let $h : X \times Y \to Z$ be a morphism. If h maps $X \times y_0$ to a single point for some $y_0 \in Y$, then there are an open neighbourhood V of y_0 and a morphism $g : V \to Z$ with $h = g \circ \mathrm{pr}_2$ on $X \times V$.*

Proof. Near $X \times y_0$ the map h lands in an affine open of Z, and a map to an affine scheme is given by functions. Since X is proper and geometrically integral, functions on $X \times V$ are functions on V. Hence h factors through V.

Let $W \subset Z$ be an affine open containing $h(X \times y_0)$. Since X is proper, the image in Y of the closed set $h^{-1}(Z \setminus W)$ is closed, and its complement contains y_0. Let V be an affine open neighbourhood of y_0 in this complement. Then h maps $X \times V$ into the affine scheme W. It is therefore determined by the ring map $\mathcal{O}(W) \to H^0(X \times V, \mathcal{O})$ [Stacks, 01I1]. By flat base change [Stacks, 02KH] and $H^0(X, \mathcal{O}_X) = k$ [Stacks, 0BUG], we have $H^0(X \times V, \mathcal{O}) = \mathcal{O}(V)$. The ring map is thus pr_2^* composed with a ring map $\mathcal{O}(W) \to \mathcal{O}(V)$, and this ring map defines g. ■

B.2 Multiplication by n

Let $\mathcal{L}$ be an invertible sheaf on an abelian variety A. We pull $\mathcal{L}$ back to $A \times A \times A$ along the partial sums of the coordinates. There the theorem of the cube gives a second order recursion for $n \mapsto [n]^*\mathcal{L}$, whose solution is

$$[n]^*\mathcal{L} \simeq \mathcal{L}^{\otimes n(n+1)/2} \otimes ([-1]^*\mathcal{L})^{\otimes n(n-1)/2}$$

(Proposition B.11). For symmetric $\mathcal{L}$ the formula gives $\mathcal{L}^{\otimes n^2}$. If $[-1]^*\mathcal{L} \simeq \mathcal{L}^{-1}$, as for an invertible sheaf of degree zero on an elliptic curve, it gives $\mathcal{L}^{\otimes n}$.

We apply the formula to a symmetric ample sheaf (Lemma B.10). It shows that $[n]$ is finite and flat of degree n^{2g}, and that $[n]$ is étale when n is invertible in k (Proposition B.12). The hypothesis on n cannot be dropped. For example, in characteristic p the differential of $[p]$ is zero, and as a consequence an elliptic curve has at most p points of order dividing p over $\bar{k}$, although $E[p]$ has rank p^2.

Definition B.8 (Abelian variety)**.** An *abelian variety* over k is a smooth proper geometrically connected k-group scheme. If A is an abelian variety, $[n]$ denotes multiplication by the integer n, and $A[n] = \ker[n]$ for $n > 0$. An invertible sheaf $\mathcal{L}$ on A is *symmetric* if $[-1]^*\mathcal{L} \simeq \mathcal{L}$.

We use three facts: commutativity, which follows from rigidity; a symmetric ample sheaf, which follows from projectivity; and the formula for $[n]^*\mathcal{L}$, which follows from the cube. Rigidity and the cube require A to be geometrically integral, which we check together with commutativity.

Lemma B.9 (Commutativity)**.** *An abelian variety A over k is geometrically integral and commutative.*

Proof. An abelian variety is geometrically reduced because it is smooth, and geometrically irreducible because the identity component of a group scheme over a field is geometrically irreducible [Stacks, 0B7R] and A is connected.

For commutativity, consider $h : A \times A \to A$, $h(x, y) = xyx^{-1}y^{-1}$. It maps $A \times \{e\}$ to e. By Lemma B.7, applied with the first factor as the proper one, there are an open $V \ni e$ and $g : V \to A$ with $h|_{A\times V} = g \circ \mathrm{pr}_2$. Composing with $(e, \mathrm{id}) : V \to A \times V$ gives $g = h(e, -) = e$, so $h = e$ on $A \times V$. To pass from $A \times V$ to $A \times A$, we compare h with the constant map e. Since $A \times A$ is smooth, hence reduced, and irreducible, the nonempty open $A \times V$ is scheme-theoretically dense. Two maps to a separated scheme that agree on a scheme-theoretically dense open agree everywhere (Lemma A.4*). Hence $h = e$ [Stacks, 056D, 01RH]. ∎

From now on we write the group law additively. Projectivity is the one structural fact that we quote.

Lemma B.10 (A symmetric ample sheaf)**.** *An abelian variety A over k is projective, and it has a symmetric ample invertible sheaf.*

Proof. Every group scheme of finite type over a field is quasi-projective [Stacks, 0BF7]. A quasi-projective proper scheme is projective [Stacks, 0B45]. If $\mathcal{L}$ is ample, then so is $[-1]^*\mathcal{L}$, since $[-1]$ is an automorphism, and the tensor product $\mathcal{N} = \mathcal{L} \otimes [-1]^*\mathcal{L}$ of two ample sheaves is ample. It is symmetric because $[-1]$ is an involution, which interchanges the two factors. ∎

Proposition B.11 (Pullback of invertible sheaves by $\{[n]\}$)**.** *Let A be an abelian variety over k and let $\mathcal{L}$ be an invertible sheaf on A. For every integer $n \geq 0$,*

$$[n]^*\mathcal{L} \simeq \mathcal{L}^{\otimes n(n+1)/2} \otimes ([-1]^*\mathcal{L})^{\otimes n(n-1)/2}.$$

In particular $[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$ if $\mathcal{N}$ is symmetric.

Proof. The idea is to produce one trivial sheaf $\mathcal{M}$ on $A \times A \times A$ by the theorem of the cube, and to pull it back along maps $A \to A \times A \times A$ chosen so that the result is a relation between $[n + 1]^*\mathcal{L}$, $[n]^*\mathcal{L}$ and $[n - 1]^*\mathcal{L}$. The theorem of the cube applies because A is geometrically integral (Lemma B.9).

Consider on $A \times A \times A$ the invertible sheaf

$$\mathcal{M} = m_{123}^*\mathcal{L} \otimes m_{12}^*\mathcal{L}^{-1} \otimes m_{13}^*\mathcal{L}^{-1} \otimes m_{23}^*\mathcal{L}^{-1} \otimes p_1^*\mathcal{L} \otimes p_2^*\mathcal{L} \otimes p_3^*\mathcal{L},$$

where the p_i are the projections and m_{12}, m_{13}, m_{23}, m_{123} are the sums of the indicated coordinates. To apply the theorem of the cube, we restrict $\mathcal{M}$ to the three slices. On $e \times A \times A$, with coordinates (x_2, x_3), the seven factors restrict to $\mathcal{L}$ pulled back along $x_2 + x_3$, x_2, x_3, $x_2 + x_3$, the constant map e, x_2 and x_3, with exponents $1, -1, -1, -1, 1, 1, 1$. Of these, the pullback along the constant map is trivial, and the others cancel. The same holds on $A \times e \times A$ and $A \times A \times e$. Hence $\mathcal{M}$ is trivial by Theorem B.6. Finally, pulling the trivial sheaf $\mathcal{M}$ back along $x \mapsto (x, x, [n - 1]x)$ for $n \geq 0$ yields

$$[n + 1]^*\mathcal{L} \otimes \mathcal{L}^{\otimes 2} \otimes [n - 1]^*\mathcal{L} \simeq [2]^*\mathcal{L} \otimes ([n]^*\mathcal{L})^{\otimes 2}.$$

To see this, note that the composites of $x \mapsto (x, x, [n - 1]x)$ with m_{123}, m_{12}, m_{13}, m_{23}, p_1, p_2, p_3 are $[n + 1]$, $[2]$, $[n]$, $[n]$, id, id, $[n - 1]$. The cases $n = 0$ and $n = 1$ of the formula are clear. For

$n = 0$, since $[0]^*\mathcal{L}$ is trivial, the relation reads $\mathcal{L}^{\otimes 3} \otimes [-1]^*\mathcal{L} \simeq [2]^*\mathcal{L}$. This is the case $n = 2$ of the formula. In general, the relation expresses $[n+1]^*\mathcal{L}$ in terms of $[n]^*\mathcal{L}$ and $[n-1]^*\mathcal{L}$, and the exponents in the formula satisfy the same second order recursion. Induction on n therefore gives the formula [Stacks, 0BFF]. In particular, for symmetric $\mathcal{N}$ the two exponents add up to n^2. ∎

For a symmetric ample $\mathcal{N}$, the degree of $[n]$ can be read off from the growth of $\chi(A, \mathcal{N}^{\otimes m})$, since pulling back by $[n]$ multiplies m by n^2 and χ grows like m^g.

Proposition B.12 (Multiplication by n)**.** *Let A be an abelian variety of dimension g over k. For every $n > 0$, the morphism $[n]$ is surjective and finite locally free of degree n^{2g}, and its kernel $A[n]$ has rank n^{2g} over k. If n is invertible in k, then $[n]$ and $A[n] \to \operatorname{Spec} k$ are finite étale.*

Proof. We use the isomorphism $[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$ twice: on curves in the fibres of $[n]$ it shows that $[n]$ is finite, and in Euler characteristics it computes the degree. In between, translation invariance shows that all fibres have the same length, which makes $[n]$ finite locally free. Finally, whether $[n]$ is étale is decided by its differential.

Since finiteness, flatness and the degree may be checked after base change to an algebraic closure, we may assume that k is algebraically closed. Choose a symmetric ample $\mathcal{N}$ (Lemma B.10); then $[n]^*\mathcal{N} \simeq \mathcal{N}^{\otimes n^2}$ by Proposition B.11. If a fibre of $[n]$ had positive dimension, it would contain a proper integral curve. Since $[n]$ maps that curve to a point, the restriction of $[n]^*\mathcal{N}$ to it is trivial, whereas the restriction of $\mathcal{N}^{\otimes n^2}$ has positive degree. Hence every fibre has dimension zero, and since $[n]$ is proper, it is finite. Its image is a closed subset of dimension g, hence equal to the integral variety A (Lemma B.9). After base change to an algebraic closure of its residue field, every fibre of $[n]$ becomes a translate of $A[n]$. Indeed, if $y = nx_0$, then translation by x_0 carries $A[n] = [n]^{-1}(0)$ isomorphically onto $[n]^{-1}(y)$, because $[n]$ is a homomorphism. Hence all fibres have the same degree $r = \dim_k \Gamma(A[n], \mathcal{O}_{A[n]})$.

To get flatness from the constancy of the fibre degree, we use the following general fact. A finite morphism f to a reduced Noetherian scheme whose fibres all have degree r is finite locally free of rank r. Indeed, near any point Nakayama's lemma gives a surjection $\mathcal{O}^{\oplus r} \to f_*\mathcal{O}$, since the fibre at that point has dimension r. This surjection is an isomorphism at each generic point, where the local ring is a field and both sides are vector spaces of dimension r. Hence on an affine open every coordinate of a section of its kernel lies in all minimal primes, and is therefore zero because the ring is reduced. Since A is smooth, hence reduced, $[n]$ is finite locally free.

Recall that for a locally free sheaf $\mathcal{E}$ of rank r on A, the function $m \mapsto \chi(A, \mathcal{E} \otimes \mathcal{N}^{\otimes m})$ is a polynomial of degree g with leading coefficient $r \deg_{\mathcal{N}}(A)/g!$. Here $\deg_{\mathcal{N}}(A) = (\mathcal{N}^g)$ is the degree of A with respect to $\mathcal{N}$, a positive integer because $\mathcal{N}$ is ample. We compare the coefficients of m^g in two expressions for $\chi(A, [n]^*\mathcal{N}^{\otimes m})$. Since $[n]$ is finite, this Euler characteristic equals $\chi(A, [n]_*[n]^*\mathcal{N}^{\otimes m})$, which by the projection formula is $\chi(A, \mathcal{N}^{\otimes m} \otimes [n]_*\mathcal{O}_A)$. As $[n]_*\mathcal{O}_A$ is locally free of rank $\deg[n]$, the coefficient is $\deg[n] \deg_{\mathcal{N}}(A)/g!$. Since $[n]^*\mathcal{N}^{\otimes m} \simeq \mathcal{N}^{\otimes n^2 m}$, it is also $n^{2g} \deg_{\mathcal{N}}(A)/g!$. Hence

$$\deg[n] \deg_{\mathcal{N}}(A) = n^{2g} \deg_{\mathcal{N}}(A).$$

Since $\deg_{\mathcal{N}}(A) > 0$, we conclude that $\deg[n] = n^{2g}$; compare [Stacks, 0BFG]. It remains to treat étaleness. The differential of the group law at $(0, 0)$ is addition of tangent vectors. By induction, the differential of $[n]$ at the identity is $n \cdot \mathrm{id}$. Since $[n]$ commutes with translations, the same holds at every point. If n is invertible, $[n]$ is therefore étale. Base change along the identity section gives the assertion for $A[n]$. ∎

It follows that the group $A(\bar{k})$ is divisible. Indeed, since $[n]$ is finite and surjective, its fibre over a point of $A(\bar{k})$ is a nonempty finite $\bar{k}$-scheme, and so it has a $\bar{k}$-point.

B.3 Tate modules

Fix a prime ℓ invertible in k. With this prime, compatible systems of ℓ-power torsion points of A form the Tate module $T_\ell A$, a $\mathbf{Z}_\ell$-module with an action of $\mathrm{Gal}(\bar{k}/k)$. For example, for a complex torus $\mathbf{C}^g/\Lambda$ it is $\Lambda \otimes \mathbf{Z}_\ell$, the first homology with ℓ-adic coefficients. In general, Proposition B.14 shows that $T_\ell A$ is free of rank $2g$ over $\mathbf{Z}_\ell$. The input is the orders of the groups $A[\ell^n](\bar{k})$, which are known from Proposition B.12, and these orders force each of these groups to be free over $\mathbf{Z}/\ell^n$ of rank $2g$. Moreover, the transition maps are surjective because $A(\bar{k})$ is divisible.

Although Frobenius can have many fixed points on the finite levels, no power of it has a nonzero fixed vector in the Tate module (Proposition B.15). For example, for the elliptic curve $E_0 : y^2 = x^3 - x$ over $\mathbf{F}_3$, the three roots of $x^3 - x$ lie in $\mathbf{F}_3$. So all of $E[2]$ is rational, and Frobenius acts trivially on it. On $T_2 E$, however, a fixed vector would be an unbounded compatible system of 2-power torsion points in the finite group $E_0(\mathbf{F}_{3^m})$. For the Jacobian of a curve, Proposition 14.3* turns this into the statement that Frobenius has no fixed class in H^1 of the curve.

Definition B.13 (Tate module)**.** Let A be an abelian variety over k and let ℓ be a prime invertible in k. The *Tate module* of A is

$$T_\ell A = \varprojlim_n A[\ell^n](\bar{k}),$$

the limit taken along the maps $[\ell] : A[\ell^{n+1}](\bar{k}) \to A[\ell^n](\bar{k})$, and we put $V_\ell A = T_\ell A \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell$.

Proposition B.14 (The integral Tate module)**.** *Let A be an abelian variety of dimension g over k, and let ℓ be a prime invertible in k. There are noncanonical isomorphisms*

$$A[\ell^n](\bar{k}) \simeq (\mathbf{Z}/\ell^n)^{2g}, \qquad T_\ell A \simeq \mathbf{Z}_\ell^{2g}.$$

The canonical projection induces an isomorphism $T_\ell A/\ell^n T_\ell A \simeq A[\ell^n](\bar{k})$. The groups $A[\ell^n](\bar{k})$, the transition maps and the projections from $T_\ell A$ are $\mathrm{Gal}(\bar{k}/k)$-equivariant, and they do not change when $\bar{k}$ is replaced by a larger algebraically closed field.

Proof. A finite abelian group of order ℓ^{2gn} killed by ℓ^n need not be free over $\mathbf{Z}/\ell^n$; for $g \geq 1$ the group $(\mathbf{Z}/\ell^2)^{2g-2} \oplus (\mathbf{Z}/\ell)^4$ has order ℓ^{4g} and is killed by ℓ^2. For $A[\ell^n]$, however, this cannot happen because we know the orders at every level together with the exact sequences given by $[\ell]$. Indeed, we use them to lift a basis of the ℓ-torsion step by step and check by counting that the lifts form a basis at each level.

The ℓ-torsion and the exact sequences. The group $A[\ell](\bar{k})$ is a vector space of dimension $2g$ over $\mathbf{F}_\ell$, since it is killed by ℓ and has order ℓ^{2g} by Proposition B.12. Moreover there are exact sequences

$$0 \longrightarrow A[\ell](\bar{k}) \longrightarrow A[\ell^{n+1}](\bar{k}) \xrightarrow{[\ell]} A[\ell^n](\bar{k}) \longrightarrow 0,$$

whose surjectivity holds because $A(\bar{k})$ is divisible: every point of $A[\ell^n](\bar{k})$ is ℓ times a point, which then lies in $A[\ell^{n+1}](\bar{k})$.

Lifting bases. Choose a basis of $A[\ell](\bar{k})$ and lift it successively along $[\ell]$; the lifts form a basis at each level. Suppose that $x_1, \ldots, x_{2g} \in A[\ell^{n+1}](\bar{k})$ lift a basis of $A[\ell^n](\bar{k})$ obtained in this way, so that $\ell^n x_1, \ldots, \ell^n x_{2g}$ is the chosen basis of $A[\ell](\bar{k})$. To see that the x_i are independent, note first that all coefficients of a relation between them are divisible by ℓ^n, as we see by applying $[\ell]$ to it.

The relation then becomes a relation between the elements $\ell^n x_i$ of the chosen basis of $A[\ell](\bar{k})$, so the coefficients are even divisible by ℓ^{n+1}. As $A[\ell^{n+1}](\bar{k})$ has order $\ell^{2g(n+1)}$, the x_i therefore form a basis of it over $\mathbf{Z}/\ell^{n+1}$. Passing to the inverse limit of these compatible bases, we obtain both isomorphisms and the description of $T_\ell A/\ell^n T_\ell A$.

Equivariance and independence of $\bar{k}$. The groups and maps in the statement commute with the action of $\mathrm{Gal}(\bar{k}/k)$, because the group schemes $A[\ell^n]$ and the maps $[\ell]$ are defined over k. In particular the projection $T_\ell A/\ell^n T_\ell A \to A[\ell^n](\bar{k})$ is equivariant. The chosen bases need not be equivariant, which is why the isomorphisms with $(\mathbf{Z}/\ell^n)^{2g}$ and $\mathbf{Z}_\ell^{2g}$ are noncanonical. Finally, the points of $A[\ell^n]$ and the multiplication maps do not change under a further algebraically closed extension, because $A[\ell^n]$ is finite étale and hence a finite disjoint union of points over $\bar{k}$. ∎

Over a finite field, Frobenius acts on $T_\ell A$, and by the last proposition this action is compatible with its action on each $A[\ell^n](\bar{k})$. The next proposition shows that no power of Frobenius has a nonzero fixed vector. For the Jacobian of a curve, this is the input to Proposition 14.3*.

Proposition B.15 (No Frobenius-fixed vector)**.** *Let $A_0/\mathbf{F}_q$ be an abelian variety, let $A = A_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let F be geometric Frobenius on $T_\ell A$. For every $m > 0$,*

$$(T_\ell A)^{F^m} = 0, \qquad (V_\ell A)^{F^m} = 0.$$

Consequently $F^m - 1$ is an automorphism of the finite-dimensional $\mathbf{Q}_\ell$-vector space $V_\ell A$, and hence of $V_\ell A \otimes_{\mathbf{Q}_\ell} E$ for every field extension E of $\mathbf{Q}_\ell$.

Proof. A fixed vector is a compatible system of torsion points, each rational over $\mathbf{F}_{q^m}$. But the rational points form a finite group, so the orders of the ℓ-power torsion points in it are bounded, and compatibility under $[\ell]$ then forces the system to vanish. The statement for $V_\ell A$ then follows by clearing denominators.

Since an automorphism and its inverse have the same fixed vectors, we may use arithmetic Frobenius instead. On $A(\overline{\mathbf{F}}_q)$ its mth power fixes exactly $A_0(\mathbf{F}_{q^m})$. This group is finite, because on each affine open of a finite affine cover a rational point is determined by finitely many coordinates in the finite field. Hence the ℓ-primary part of $A_0(\mathbf{F}_{q^m})$ is killed by some ℓ^e. Let (t_n) be a fixed compatible sequence. Each t_{n+e} lies in this ℓ-primary part, and since the transition maps are $[\ell]$, we find

$$t_n = \ell^e t_{n+e} = 0$$

for every n. This proves that $(T_\ell A)^{F^m} = 0$.

Since $T_\ell A$ is free by Proposition B.14, it injects into $V_\ell A$. A suitable power of ℓ times a fixed vector of $V_\ell A$ is a fixed vector of $T_\ell A$, and so it vanishes. Hence $(V_\ell A)^{F^m} = 0$. In other words, $F^m - 1$ is injective on the finite-dimensional space $V_\ell A$, and therefore bijective. ∎

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References marked * are to parts of the blueprint this document leaves out:

Proposition 14.3 No fixed class in degree one, in III Weights and the Riemann hypothesis (`proposition-no-fixed-class-degree-one`)

Lemma A.4 Uniqueness of extensions from a dense open, in Appendices (`lemma-uniqueness-maps-dense-open`)

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